Container-based High-Performance Computing Approaches for Simulating Large-Scale Biological Neural Networks

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Introduction

1. Modeling large-scale, biophysically detailed neural networks increasingly relies on high-performance computing (HPC).

2. However, deploying simulations of such networks across diverse HPC environments remains challenging due to software incompatibilities and complex dependencies.

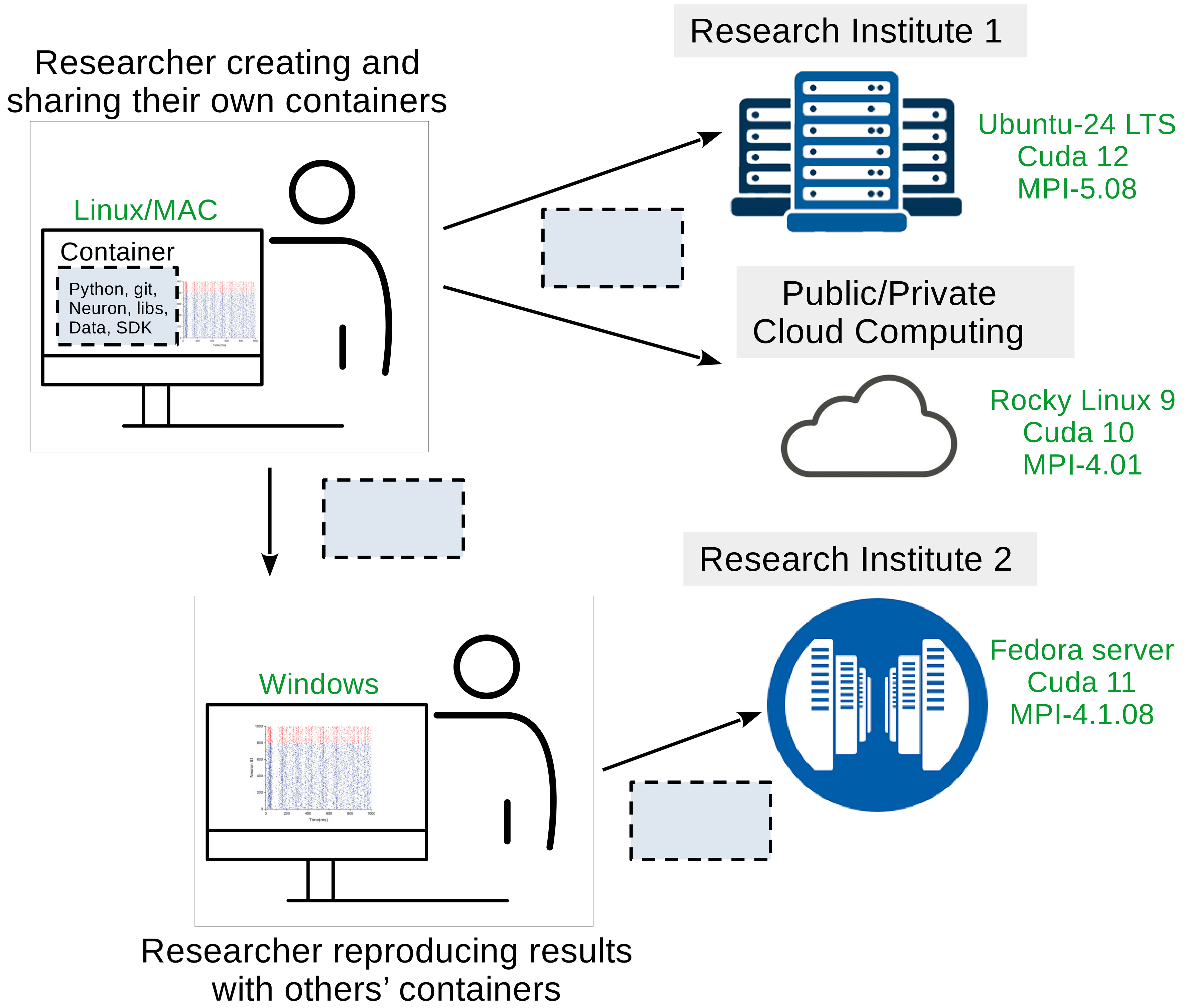
3. We address this issue using singularity containers that encapsulats the entire simulation environment, enabling neuroscientists to achieve portability and reproducibility without requiring root privileges on HPC/Cloud systems.

4. We give a practical demonstrations by benchmarking HPC environments with cerebellar mossy fiber–granule cell network simulations.

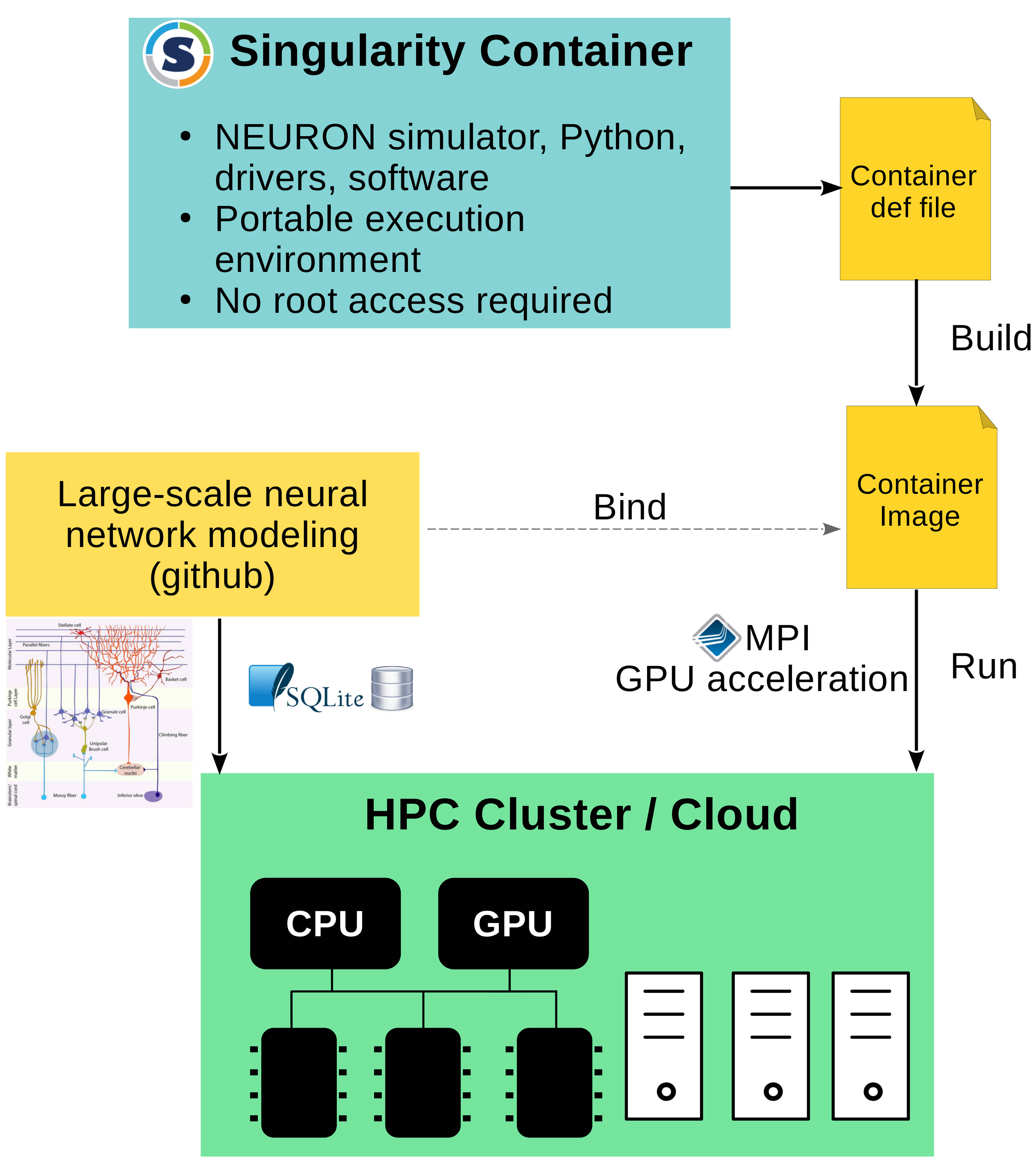
Containerization for Scientific Computing

Container is a lightweight, standalone, and executable package.

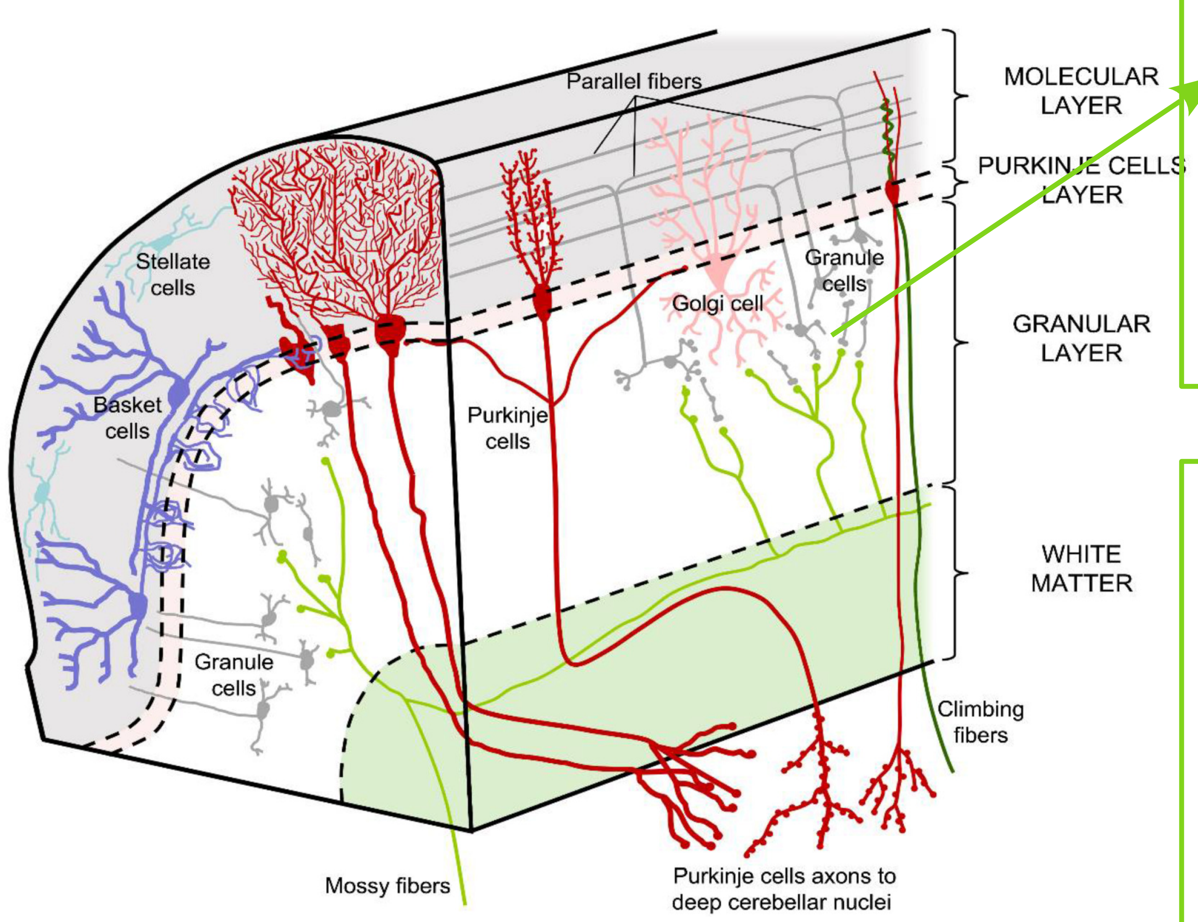
It is designged to include everything needed to run a scientific software.



Our Computational Framework



Benchmarking Data and Network: Cerebellar Mossy Fiber → Granule Cell



EM Data

MF - GRC Connectivity

Nguyen, T.M., Thomas, L.A., Rhoades, J.L. et al. Structured cerebellar connectivity supports resilient pattern separation. **Nature** 613, 543–549 (2023)

MF 1042 cells
GRC 3953 cells

Cell Models

MF → spike generators
GRC → Biophysically detailed model with 12 ion channel mechanisms

Sudhakar, S.K., Hong, S., Raikov, I., Publio, R., Lang, C., Close, T., Guo, D., Negrello, M., De Schutter, E., 2017. Spatiotemporal network coding of physiological mossy fiber inputs by the cerebellar granular layer. **PLOS Comput Biol** 13

We run multiple copies ($n = 80$) of the network connectivity model simultaneously to simulate extreme computational workloads.

This allows us to rigorously assess the scalability and performance limits of GPUs under high-demand conditions.

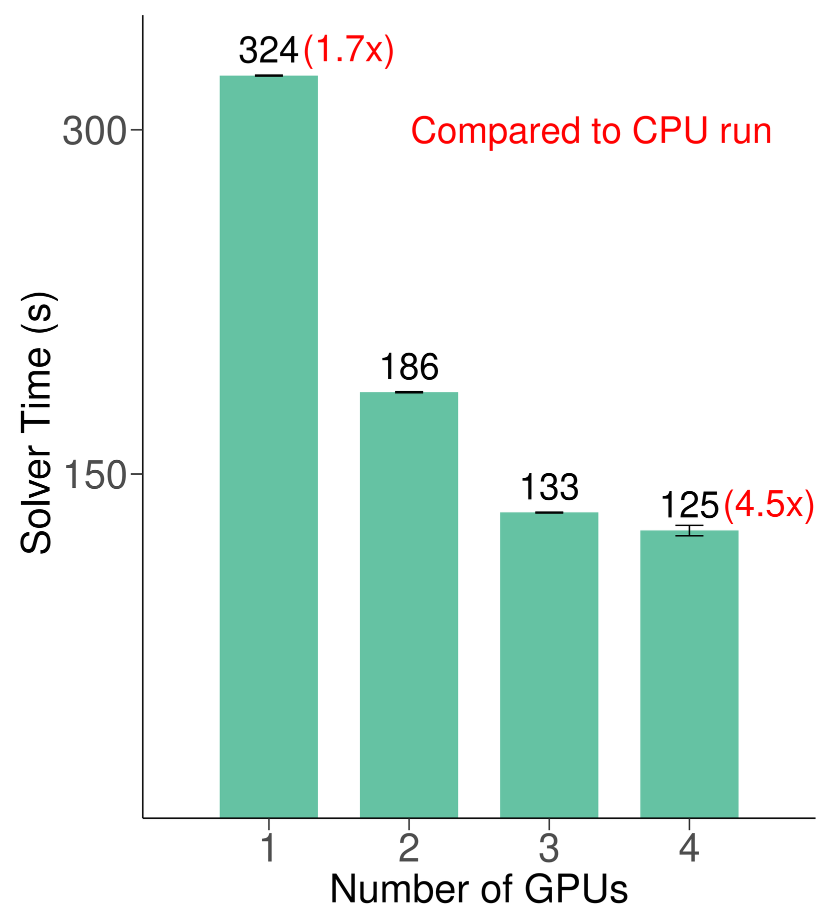
Table: Network Configuration and Simulation Parameters

Number of MF	85,600
Number of GRC	314,000
Number of compartments	942,000
Number of synapses	990,960
Simulation time	2 s
Simulation software	COREneuron

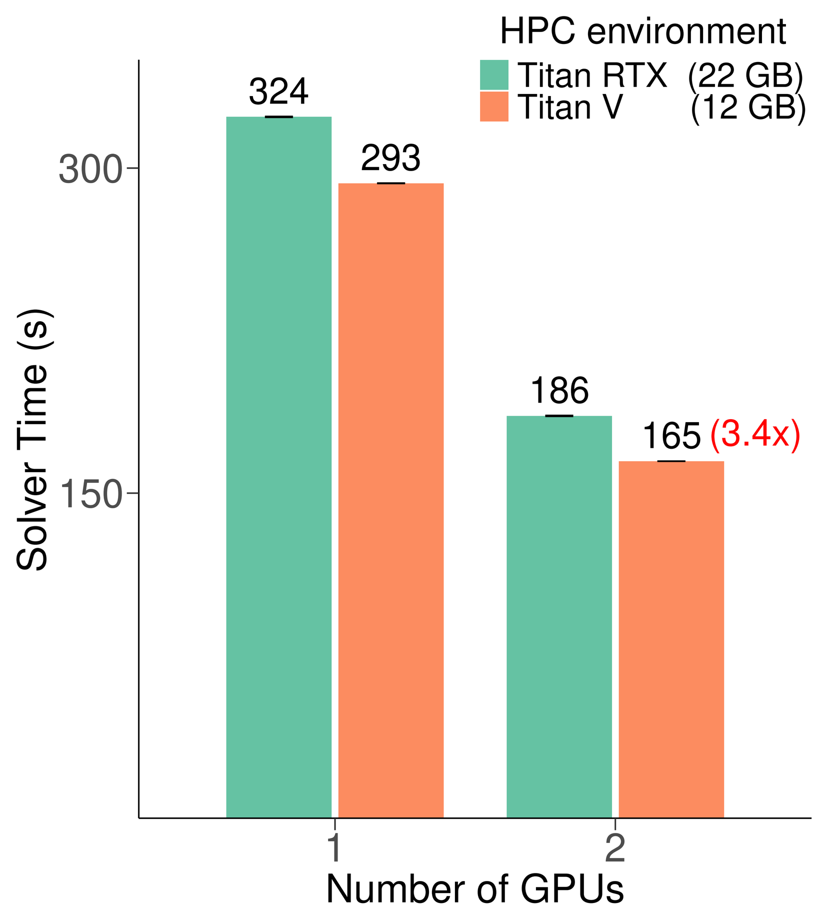
Benchmarking HPC Environments

Table: Performance in CPU parallel computing (30-task MPI)

CPU	Core	Solver Time (s)	Simulator
Intel(R)-Xeon	30	564	CORENeuron-CPU



(a) TITAN RTX architecture: Comparison of solver time for varying numbers of GPUs in CORENeuron-GPU simulator



(b) Comparison of solver time between HPC environments

Figure: Solver time comparisons in CORENeuron simulations across HPC platforms.

TITAN RTX outperform 30-core CPUs by up to 4.5x, with performance improving as GPU count increases.

Future works:

☞ To include more realistic model of granule layer, including Golgi cells.

☞ To extend the model to include Purkinje cells and other interneurons for a complete cerebellar circuit.

Acknowledgment

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